

Model	Environment Type												Memory Req.				Env. Dynamics				Overall	
	Maze		Shooter		Control		Dodge		Collect		Sport		Single		Multiple		Static		Dynamic		All	
	S	P%	S	P%	S	P%	S	P%	S	P%	S	P%	S	P%	S	P%	S	P%	S	P%	S	P%
Closed-source Models																						
Seed-2.1-Pro	64.4	70	43.3	49	88.4	73	47.1	51	52.3	60	54.3	80	68.9	70	52.7	57	74.3	70	47.0	54	54.4	58
Gemini-3.6-Flash	61.0	70	37.9	39	88.1	80	39.9	44	42.2	63	29.7	50	67.1	70	47.3	52	77.2	75	39.2	45	49.5	53
Gemini-3.1-Pro	49.1	54	44.9	46	65.5	53	42.0	48	40.3	45	34.3	70	61.5	70	46.4	51	67.7	69	40.8	47	48.0	53
GPT-5.6-sol	52.0	53	37.6	43	82.7	87	38.1	43	41.5	43	35.0	60	82.5	85	42.4	47	68.3	67	38.8	46	46.7	51
Gemini-3-Flash	36.9	37	37.6	39	64.5	67	34.7	36	31.1	40	34.3	60	52.2	50	38.1	41	54.7	53	34.0	38	39.6	42
Seed-2.0-Lite	39.0	51	26.5	24	86.3	80	31.4	36	36.2	48	32.0	60	54.2	60	34.9	39	50.6	49	31.9	38	37.0	41
Open-weight Models																						
DeepSeek-V4.1-Flash	34.1	36	28.7	26	71.6	67	32.4	33	26.0	30	32.3	50	46.1	45	33.5	34	51.0	50	28.8	30	34.8	35
Kimi-K3	32.3	40	26.4	34	70.1	73	27.2	34	25.4	30	26.3	40	51.8	55	30.2	37	46.7	46	27.3	36	32.5	39
Qwen3.5-397B-A17B	28.2	33	30.3	27	73.6	67	28.1	27	25.4	30	25.7	40	44.2	45	31.1	30	45.0	42	27.9	27	32.5	31
DeepSeek-V4F-V	21.0	29	26.0	34	47.1	53	24.1	30	19.3	30	21.7	30	26.6	15	25.0	32	30.1	26	23.4	32	25.2	30
gemma-4-31B-it	20.4	28	17.7	29	32.8	27	19.7	25	14.4	18	13.7	10	40.1	35	19.7	24	29.8	25	18.9	25	21.9	25
MiMo-v2.5	20.0	21	18.3	26	34.6	27	21.4	27	13.1	28	16.0	20	25.6	15	19.8	24	22.2	14	19.8	26	20.4	23
Qwen2.5-VL-72B	14.3	16	16.3	35	20.1	7	20.4	31	11.5	25	10.3	0	19.0	10	17.5	25	16.9	8	17.9	29	17.7	23
MiMo-VL-7B	13.9	15	15.8	10	19.7	13	19.9	16	13.3	10	12.7	10	20.6	20	17.0	13	17.1	14	17.5	14	17.4	14
Qwen3.6-27B	10.5	18	13.8	36	17.5	7	19.2	33	12.2	28	22.7	40	17.3	10	16.8	29	14.8	8	17.6	34	16.9	27
GLM-4.5V	10.9	15	13.1	27	28.7	27	18.1	26	11.9	15	12.0	10	17.2	5	16.5	23	16.6	10	16.6	25	16.6	21
Qwen3.5-27B	13.3	15	12.4	35	22.4	13	17.8	32	12.2	20	20.0	40	18.5	0	16.2	27	14.2	4	17.3	32	16.4	24
Qwen3-Omni-30B	9.7	13	5.8	20	19.5	13	15.6	27	5.3	18	16.7	30	19.6	10	12.8	22	16.9	10	12.3	24	13.6	20
Qwen3-VL-235B	12.5	6	6.9	3	19.7	13	11.7	5	11.6	5	15.3	10	21.6	5	11.2	5	14.2	6	11.6	5	12.3	5
Reference Baselines																						
Random	7.9	10	19.1	23	16.8	13	15.5	15	15.0	15	19.4	30	5.6	2	15.0	15	8.4	5	16.1	17	14.0	14
Human-score	92.1	96	76.4	85	94.6	89	82.8	90	91.3	99	69.8	97	99.0	100	81.8	90	98.1	99	78.3	88	83.6	91

Main results on V-ICAL. **S**: normalized score (0–100); **P%**: pass rate. **MEMORY REQ.**: memory requirement for action selection. Single means that the current frame is sufficient; Multiple means the agent must retain one or more earlier post-action observations and integrate them to choose a later action. **ENV. DYNAMICS**: whether the environment changes independently of the agent (Static vs. Dynamic).